

Automotive Transmitter Triggered by a Headlight Pulse

Battery-free gate-opener module — powered by the vehicle high beam — 433.92 MHz RF transmitter with EETX433A

Application guide for an in-vehicle transmitter that opens gates and barriers with a flash of the high beam. Powered directly from the high-beam lamp (12/24 V), it needs no battery and never leaves the vehicle.

FREQUENCY

433.92 MHz

RF POWER

13 dBm

POWER

Headlight 12 / 24 V

BATTERY

None

VERSION

v1.0

BASED ON IC

EETX433A

DEVELOPED BY

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DATE

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• Introduction

This application note describes how to build an **automotive gate-opener transmitter** triggered by a **high-beam pulse**, using the **EETX433A** multifunction transmitter. Instead of a battery-powered handheld remote, the module is wired to the high-beam lamp circuit and powered by the same voltage that lights the lamp. A flash of the high beam near the gate transmits the RF code and opens it.

The EETX433A integrates a 433.92 MHz PLL oscillator, OOK/ASK modulator, power amplifier and the encoding logic in a single SOP-8 package. An input stage (rectifier bridge + 3.3 V LDO regulator) adapts the vehicle voltage (12 V in cars and motorcycles, 24 V in trucks), tolerates either connection polarity and dispenses with internal batteries.

The product goal is a transmitter that **never leaves the vehicle** and **never needs a battery change**. The IC electrical details (pinout, absolute ratings, operating conditions, RF frame format and code mapping) are given in the EETX433A datasheet and referenced throughout this document.

• Operating Principle

The module input is wired to the high-beam circuit: the **positive** wire to the high-beam feed (fuse/positive of the headlight) and the **negative** wire to the vehicle chassis ground. When the driver flashes the high beam, the lamp voltage appears at the input, powers the module and triggers the transmission.

The input stage makes the circuit robust in the automotive environment: the **rectifier bridge** makes the module immune to connection polarity and rectifies the pulse; the **3.3 V LDO regulator** converts the 12/24 V from the headlight into the EETX433A operating voltage, with filter capacitors against spikes and noise from the vehicle electrical system. The presence of the high-beam voltage generates a rising edge (0→1) on the **SENIN** pin, which triggers the RF code (the same trigger mechanism used by the magnetic sensor modes).

A brief flash (recommended to hold for ≥ 2 s) guarantees the complete transmission: wake-up (~50 ms) + PLL settling (~0.2 ms) + the RF packets. If the high beam is left **on continuously** (e.g. night driving), transmission is **automatically cut after ~5 s**, avoiding continuous emission; to transmit again, turn the high beam off and flash again. With the headlight off, the module is completely unpowered — **zero drain** on the vehicle battery.

• Wiring Scheme

The module is wired in parallel with the high-beam lamp and powered by it. The input stage protects the EETX433A and adapts the vehicle voltage:

- **Input (J1):** two wires — the positive goes to the high-beam circuit (fuse/positive of the headlight) and the negative to the chassis ground.
- **Rectifier bridge:** lets the module work regardless of connection polarity and rectifies the headlight pulse.
- **3.3 V LDO regulator:** converts the 12/24 V from the headlight into the EETX433A operating voltage, with filter capacitors for spikes and noise from the vehicle electrical system.
- **Trigger:** the presence of the high-beam voltage drives the SENIN pin high (0→1 edge), triggering transmission of the code.

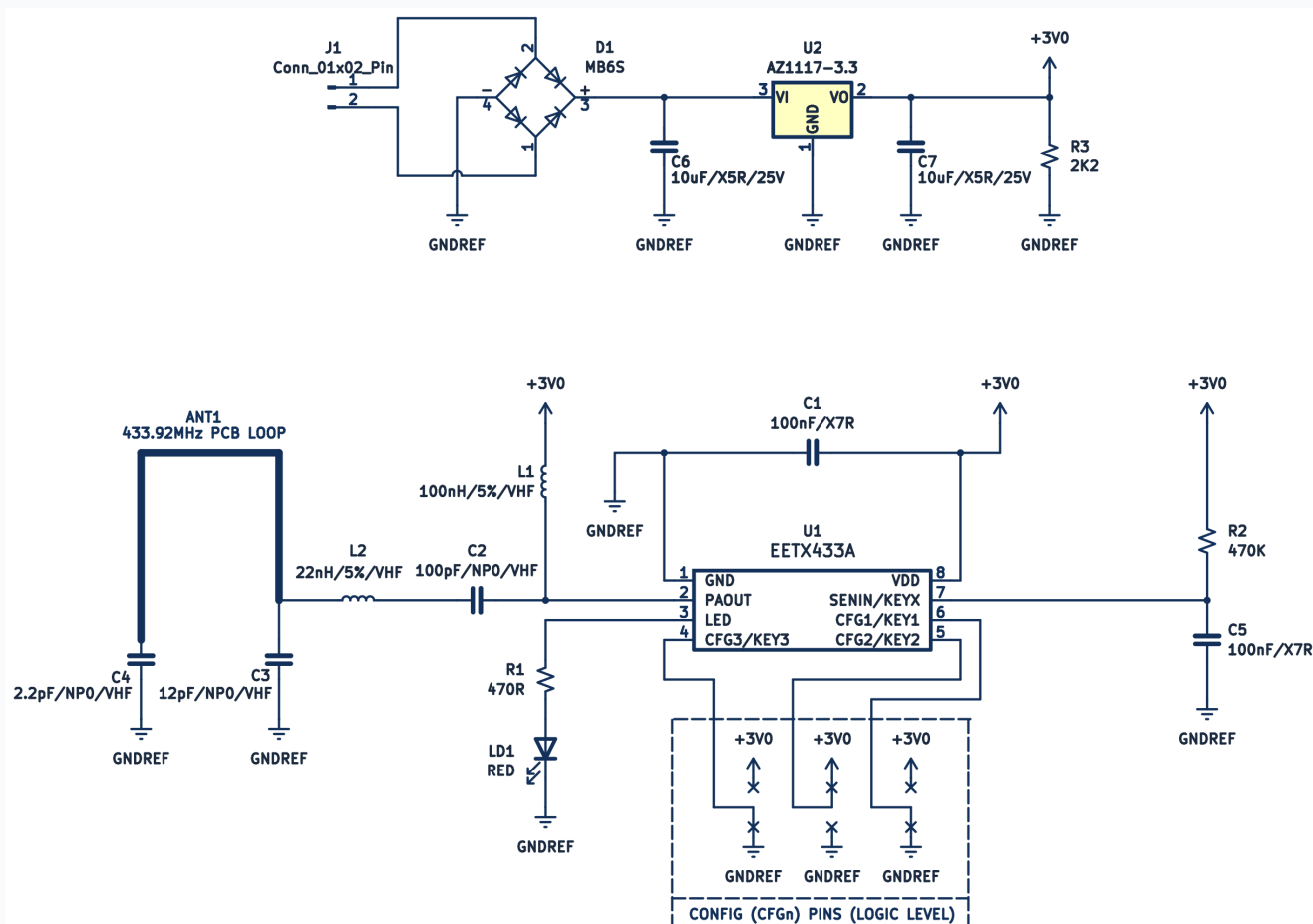


Figure 1: Transmitter powered by the high beam — rectifier bridge + 3.3 V LDO + EETX433A (433.92 MHz PCB antenna)

• Configuration (CFG Pins)

The transmitter uses the same configuration as a magnetic sensor: the CFG pins select the code and the SENIN pin is the trigger input (here, the presence of the high beam). Since there is a single event — the trigger — the Smart mode (open/close) is not used; only the violation modes:

MODE	FUNCTION	CFG3	CFG2	CFG1	SENIN
1	Trigger (Violation)	0	0	1	Trigger
2	Trigger (Violation)	0	1	0	Trigger
6	Trigger (Violation)	1	1	0	Trigger

The CFG pins are sampled only at power-up (at each flash, when the module is energized). The code is set at the factory. Mode 3 (Smart open/close) and low-battery signaling do not apply — the module is vehicle-powered and has no battery.

• Transmitted Events and Codes

At each flash, the 4-bit data field (D3D2D1D0) transmits the trigger code according to the configured mode:

MODE	FUNCTION	Event (flash)	Code (4 bits)
1	Violation	Trigger	0001 (1)
2	Violation	Trigger	0011 (3)
6	Violation	Trigger	0010 (2)

The code differs between modes, allowing distinct units to be pre-programmed to trigger different receivers/gates. The unique 20-bit address identifies each module at the receiver. There is no closing event and no low-battery event in this application.

• Operating Sequence

The sequence starts when the high beam is switched on and the module is powered. The LED indicates transmission. There is no need to hold the high beam on — one pulse is enough.

Phase	Condition	Behavior	LED
Idle	High beam off	Module unpowered — zero drain on the vehicle battery	Off
Power-up	High-beam flash	Bridge + LDO supply 3.3 V; wake-up ~50 ms + PLL ~0.2 ms	—
Transmission	High beam present · SENIN 0→1	Sends the trigger code; hold the flash ≥ 2 s for the full packet	Blinks on transmit
Auto cutoff	High beam continuous	Transmission stops after ~5 s to avoid continuous emission	Off after cutoff
Re-arm	Turn off and flash again	Switch the high beam off and flash again to transmit once more	—

PRACTICAL USE

As you approach the gate, flash the high beam (~2 s) to trigger — a single pulse is enough. If you drive with the high beam continuously on, the module cuts transmission after ~5 s, so the transmitter is not emitting while you drive. For a new trigger, turn off and flash again.

- **Transmitted RF Signal**

Transmission follows the EETX433A packet format: pulse-width OOK encoding at 433.92 MHz, basic time unit $\lambda = 500 \mu\text{s}$ (each bit takes $3\lambda = 1.5 \text{ ms}$). Each packet carries pilot, start, 20 bits of programmable address (factory-unique), 4 bits of data and 4 bits of fixed anti-code, totaling $108\lambda \approx 54 \text{ ms}$, repeated while the trigger is active, up to the $\sim 5 \text{ s}$ limit.

In the headlight trigger, the 4-bit data field carries the trigger code of the configured mode (see Events and Codes). The 20-bit address identifies the module at the gate receiver. See the EETX433A datasheet for the full frame structure and code mapping.

• Technical Characteristics

Values for the complete transmitter module (EETX433A IC + input stage). Power comes from the high-beam lamp; there is no battery.

Parameter	Value
Operating frequency	433.92 MHz
Channels	1
Modulation	OOK / ASK
RF power (typ.)	13 dBm
Input power	High beam 12 V (cars/motorcycles) / 24 V (trucks)
Polarity immunity	Rectifier bridge (any polarity)
Internal regulation	3.3 V LDO
Battery	None – powered by the headlight
Idle consumption	Zero (unpowered with the headlight off)
Transmit current	12 mA @ 3.3 V
Trigger	Presence of the high beam (0→1 edge on SENIN)
Recommended flash	≥ 2 s (full transmission)
Transmission cutoff	≈ 5 s (continuous high beam)
Available modes	1, 2, 6 (violation)
IC package	SOP-8

IC RF per the EETX433A datasheet (13 dBm). The input stage (bridge + LDO) adapts the headlight voltage and protects the EETX433A against reverse polarity and transients from the vehicle electrical system.

• Installation and Guidelines

- **Electrical connection:** connect the module's positive wire to the high-beam circuit (fuse/positive of the headlight) and the negative wire to the chassis ground, as close as possible. A high-beam flash should power the module.
- **Fuse and safety:** respect the vehicle headlight wiring and fuse and do not overload the circuit. Installation must follow the vehicle manufacturer's electrical rules.
- **Maintenance-free:** the module has no battery and stays in the vehicle — it never needs recharging or a battery change.
- **Coding:** each module has a factory-unique address; learn it into the gate receiver. The same module can be enrolled in more than one gate (compatible receivers on the same frequency).
- **Operation:** approach the gate and flash the high beam. Avoid driving with the high beam continuously on near the gate; transmission is cut after ~5 s.
- **Antenna:** use the printed 433.92 MHz loop antenna per the reference schematic and the datasheet; position the module avoiding large metal masses over the antenna.

• Soldering and Reflow

Excess temperature can damage the EETX433A, causing thermal deformation of the package and a shift of the 433.92 MHz frequency. The reflow limit is **≤ 210 °C for ≤ 10 s** — the component's own maximum rating, more restrictive than the standard J-STD-020 reflow classifications (220–235 °C for Sn-Pb).

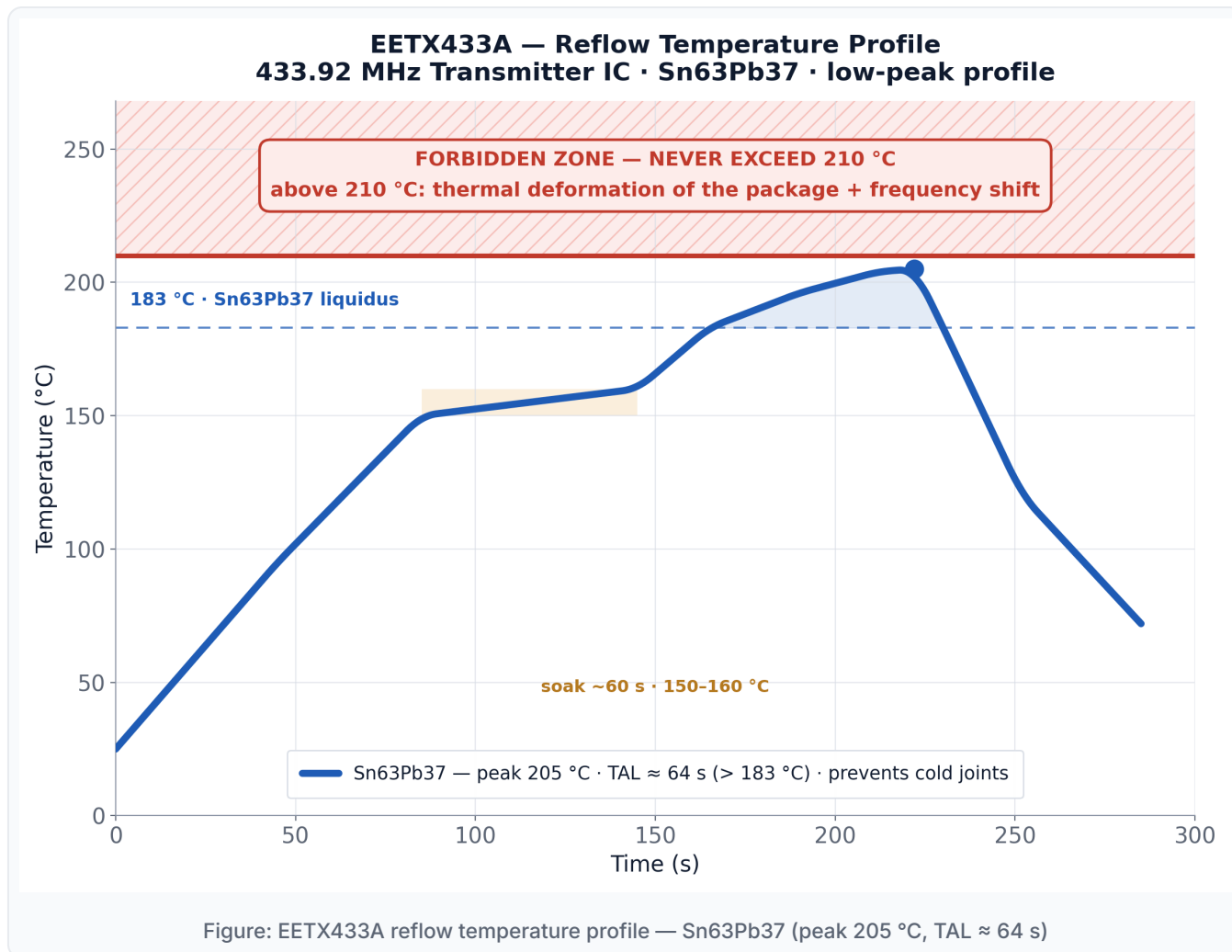
Alloy: **Sn63Pb37** (eutectic, 183 °C liquidus — the market standard), whose reflow peaks typically at **205–220 °C**; for the EETX433A the peak stays at the low end: board peak **205–210 °C** (target ~205 °C, never above 210 °C), compensated by a **TAL of 60–75 s**. Because the peak is modest, the time above liquidus is extended — and it is this combination that forms the intermetallic without exceeding the component limit.

To avoid **cold joints**: keep the soak of ~60 s between 150–160 °C (activates the flux and evens the board) and the **TAL of ~64 s above 183 °C** (ensures full melt and good wetting). Ramp-up ≤ 2.5 °C/s and cooling ≤ 4 °C/s. A high, short peak is not needed: a low peak plus an adequate TAL produces the same joint, safely for the IC.

NARROW WINDOW — MEASUREMENT IS MANDATORY

The Sn63Pb37 window here is only ~5 °C wide: **below 205 °C** there is a risk of cold joints; **above 210 °C** the IC is damaged (SOP-8 deformation and frequency drift). The profile **must be measured with a thermocouple on the board**, at least once per line. NEVER exceed 210 °C.

• Soldering and Reflow



Phase	Time (s)	Temperature (°C)
Ramp-up	0–85	25 → 150
Preheat (soak)	85–145	150 → 160
Ramp to peak	145–212	160 → 204
Peak (reflow)	212–228	205 (max. 210)
TAL — time above liquidus	≈ 64	> 183
Cooling	228–285	205 → 72

Sn63Pb37 (eutectic, 183 °C liquidus) — market standard. Low-peak profile: peak 205–210 °C (never above 210 °C), compensated by a TAL of 60–75 s. A ~60 s soak at 150–160 °C and a ~64 s TAL prevent cold joints; ramp-up ~1–2.5 °C/s · cooling ≤ 4 °C/s · max. 2 reflow cycles. Hand soldering (samples): tip ≤ 300 °C, contact ≤ 5 s.

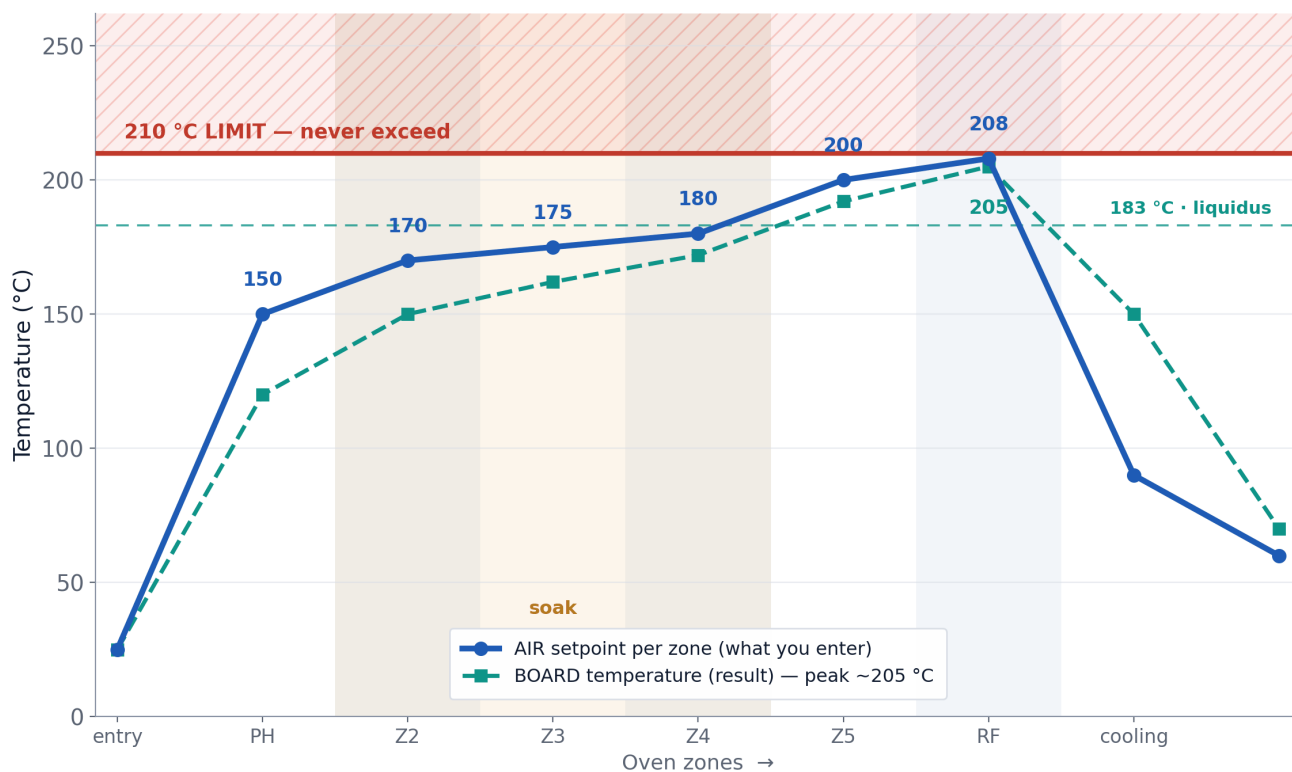
• Practical oven setup — zone profile

The chart below shows the reference setup for a tunnel oven: the **air setpoint per zone** (solid line — 150 · 170 · 175 · 180 · 200 · 208 °C) and the resulting **board temperature** (dashed). Air setpoints are a starting point — tune by profiling with a thermocouple on the board. What matters is the board result: **peak ~205 °C and TAL ~64 s**, never above 210 °C.

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EETX433A — Zone profile in straight lines
Sn63Pb37 · entered air setpoint × resulting board temperature



Zone profile — air setpoint × board temperature (peak ~205 °C · TAL ~64 s · ≤ 210 °C).

To learn more about the EETX433A and obtain the full datasheet, visit: engeletron.com.br

Legal Notice

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